Noncommutative homotopy theory

Ulrich Bunke*

October 12, 2026

Abstract

Contents

1 Overview **2**
1.1 The goals of the course 2

1 Overview

1.1 The goals of the course

In this course, we will study C^*-algebras using homotopy theoretic methods. This will allow us to see basic principles of homotopy theory in action in the setting of noncommutative topology. In particular we will see a realization of motivic methods in a specific example.

Remark 1.1. In the following we list parallel examples from various fields with very similar general structure:

*Fakultät für Mathematik, Universität Regensburg, 93040 Regensburg, ulrich.bunke@mathematik.uni-regensburg.de

field	theories	example	motives
alg. Toplology	Homology theories	$H_{\mathtt{sing}}$	category of spectra $\mathbf{Sp}$
alg. geometry	Weil homology theories	H_{et}	stable mot. homot. cat. HS
coarse geometry	coarse homology theories	$K\mathcal{X}$	coarse motivic $\mathbf{Sp}\mathcal{X}$
stable ∞-categories	localizing invariants	K^{alg}	univ. localizing invariant $\mathcal{U}_{\mathrm{loc}}$
C^*-algebras	homological functors	K^{top}	KK and E-theory

In the following, we indicate the meaning of the word *noncommutative*. Geometric objects are often studied through their algebras of functions: regular functions in algebraic geometry, smooth functions in differential geometry, and continuous functions in topology. These algebras are commutative. Noncommutative geometry extends constructions and invariants from commutative to general associative algebras which are interpreted as representatives of *noncommutative spaces*.

A further principle is Morita invariance: a construction or an invariant should depend on an appropriate category of modules over an algebra rather than on that algebra itself. This principle leads to modelling spaces through appropriate categories of modules and their generalizations, and to phrase constructions and invariants using categorical language.

We will study an instance of this idea in topology. Gelfand–Naimark duality identifies compact Hausdorff spaces with commutative unital C^*-algebras, contravariantly, via their continuous functions. The category of commutative C^*-algebras embeds into the category of all C^*-algebras whose objects are henceforth interpreted as noncommutative spaces. Morita invariance comes in through $\mathbb{K}$-stability: constructions and invariants should be invariant under tensoring with the C^*-algebra of compact operators.

Example 1.2. Let X be a finite set and G be a finite group acting on X. We compare the colimit of the G-action in commutative topology with a noncommutative version in which equality is weakened to unitary conjugacy.

The data gives the algebra $C(X) := \bigoplus_X \mathbb{C}$ generated by the components δ_x for x in X with the action of G on the index set. In topology, we would consider the fixed point set X^G and the inclusion $X^G \to X$ inducing the projection $C(X) \to C(X^G)$. Since Gelfand duality is contravariant, it switches limits and colimits and we would get $C(X)/G \cong C(X^G)$. This makes δ_x equal to δ_{gx}. If for example the action of G on X is free, then

$$C(X^G) \cong C(X)/G \cong C(\emptyset) \cong 0 \ .$$

If $X = *$, then

$$C(X^G) \cong C(X)/G \cong C(X) \ .$$

The proposal of non-commutative topology is not to make δ_x equal to δ_{gx}, but to add a unitary ρ_g for every g in G such that $\rho_g \delta_x \rho_g^{-1} = \delta_{gx}$ and $\rho_g \rho_h = \rho_{gh}$. The algebra $C(X) \rtimes G$ generated by $C(X)$ and the ρ_g for all g in G subject to these relations is called the crossed product. It retains more information than $C(X^G)$.

For example, if the action is free, then

$$C(X) \rtimes G \cong \bigoplus_{X/G} \mathtt{Mat}_{|G|}(\mathbb{C}) \ .$$

If $X = *$, then $C(*) \rtimes G \cong C^*(G) \cong \bigoplus_{\rho \in \hat{G}} \mathtt{Mat}_{\dim(\rho)}(\mathbb{C})$, where $\hat{G}$ is the dual of G, the set of isomorphism classes of finite-dimensional representations. The following schematically pictures what we have done above:

commutative colimit	add unitaries ρ_g
force $g(a) = a$	$g(a) = \rho_g a \rho_g^*$
$\Downarrow$	$\Downarrow$
$C(X^G)$	$C(X) \rtimes G$
colimit in $*\mathbf{Alg}$	colimit in $C^*\mathbf{Cat}[W^{-1}]$

The last line indicates the categorical setup where one can interpret adding the unitaries as taking a categorical quotient. In the case above, this happens in the 2-category $C^*\mathbf{Cat}[W^{-1}]$ which is obtained from the 1-category $C^*\mathbf{Cat}$ of C^*-categories by inverting unitary equivalences. □

This leads us to explain the general principles behind the notion of *homotopy theory*. Typically one starts with a category $\mathcal{C}$ and chooses a class of morphisms W, called *weak equivalences*. One then universally inverts them by a Dwyer–Kan localization,

$$\ell : \mathcal{C} \longrightarrow \mathcal{C}[W^{-1}]$$

whose precise definition requires the language of ∞-categories. Unlike an ordinary localization, it retains mapping spaces and their higher homotopies.

Example 1.3. For example, localizing the category $\mathbf{CW}$ of CW-complexes at homotopy equivalences gives the ∞-category $\mathbf{Ani}$ of *anima*, or homotopy types. □

Doing homotopy theory is then understood as studying constructions with objects from $\mathcal{C}$ and their interpretation in $\mathcal{C}[W^{-1}]$. Typical examples are homotopy limits or colimits.

Example 1.4. Let $A : I \to \mathcal{C}$ be a diagram. A cone $\tilde{A} : I^{\triangleright} \to \mathcal{C}$ with tip $A(\infty)$ is called *homotopy colimit cone*, if the $\ell(\tilde{A})$ is a colimit cone in $\mathcal{C}[W^{-1}]$, i.e.,

$$\ell A(\infty) \simeq \operatorname*{\mathtt{colim}}_I \ell A \ .$$

So a homotopy colimit is data in $\mathcal{C}$ which presents a colimit in the localization.

The same reasoning applies to homotopy limits. □

Example 1.5. Let $\mathbf{Cat}_{\mathbb{C}}$ be the 1-category of $\mathbb{C}$-linear categories and linear functors. The localization

$$\ell : \mathbf{Cat}_{\mathbb{C}} \to \mathbf{Cat}_{\mathbb{C}}[W^{-1}]$$

at the equivalences produces the 2-category of $\mathbb{C}$-linear categories, functors and isomorphisms.

A unital algebra A can be viewed as a $\mathbb{C}$-linear category with a single object. If a group G acts on A, then the (algebraic) crossed product fits into

$$\ell(A \rtimes G) \simeq \operatorname*{colim}_{BG} \ell(A) \ .$$

So $A \rtimes G$ is the homotopy colimit of A with respect to the G-action. □

In the course, we will introduce the language of ∞-categories as it is needed. We emphasize the applications and take the internal theory as granted.

In this course we will start from the category $C^*\mathbf{Alg}^{\mathrm{nu}}$ of C^*-algebras. Different classes of weak equivalences in $C^*\mathbf{Alg}^{\mathrm{nu}}$ lead to different homotopy theories. Our principal examples will be $\mathbb{K}$-stabilized homotopy theory, which we call Morita theory, and the categories KK and E representing KK- and E-theory. Understanding their categorical properties will require both homotopy theoretic methods and results about C^*-algebras. We will develop these ingredients step by step.

We start the course with an introduction of the category $C^*\mathbf{Alg}^{\mathrm{nu}}$ of complex, possibly non-unital C^*-algebras and possibly non-unital homomorphisms in a way which makes its categorical properties visible as quickly as possible.

Under Gelfand duality, its full subcategory of commutative algebras is equivalent to the category $(\mathrm{LCH}^+)^{\mathrm{op}}$, where LCH^+ denotes locally compact Hausdorff spaces and proper continuous maps defined on open subsets. The inclusion

$$\mathrm{LCH}^+ \hookrightarrow C^*\mathbf{Alg}^{\mathrm{nu}\,\mathrm{op}}$$

is the extension of the category of classical spaces by non-commutative spaces.

The main motivating invariant of non-commutative spaces is topological K-theory,

$$K : C^*\mathbf{Alg}^{\mathrm{nu}} \longrightarrow \mathbf{Sp},$$

where $\mathbf{Sp}$ is the ∞-category of spectra. It extends the classical topological K-theory of locally compact Hausdorff spaces. Its basic properties are homotopy invariance, $\mathbb{K}$-stability, exactness, and compatibility with filtered colimits. They motivate the introduction of the localizations

$$\mathrm{kk} : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathrm{KK} \ , \qquad \mathrm{e} : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathrm{E}$$

implementing these properties universally. The difference between kk and e lies the details of the exactness condition.

The categories KK and E capture the motives of noncommutative spaces.

Essentially by definition, we obtain

$$K(A) \simeq \mathrm{map}_{\mathrm{KK}}(\mathrm{kk}(\mathbb{C}), \mathrm{kk}(A)) \simeq \mathrm{map}_{\mathrm{E}}(\mathrm{e}(\mathbb{C}), \mathrm{e}(A)).$$

Here map denotes the mapping *spectrum*. We will use this definition of K-theory to derive its properties and to compute the groups $K_i(A) := \pi_i K(A)$ in examples.

The following subsections outline the steps towards these constructions which will be the guiding principle for this course. Theorems stated without proof here will, of course, be established later in the course.